
Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value

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Team ID: SWTID-2026-2214

PHASE 2 – REQUIREMENT ANALYSIS

1. Introduction

Requirement Analysis is the process of identifying the functional and technical requirements needed to develop the project.

The main requirement of this project is to implement a secure access-control system in ServiceNow that restricts users from accessing records based on their roles and branch values.

2. Functional Requirements

The system should provide the following functionalities:

- Create a custom Institution Details table.
- Store student and faculty information.
- Store the branch of each student as EEE, ECE, or CSE.
- Create different user roles for controlling access.
- Allow authorized users to read permitted records.
- Restrict unauthorized users from viewing records.
- Allow authorized users to create records.
- Allow authorized users to modify records.

- Allow authorized users to delete records.
 - Provide administrators with complete access.
 - Use ACL scripts to implement customized access rules.
 - Test the access permissions using different users and roles.
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3. Non-Functional Requirements

The system should satisfy the following requirements:

- **Security:** Unauthorized users must not access restricted records.
 - **Reliability:** ACL rules should work consistently.
 - **Maintainability:** Access rules should be easy to update.
 - **Scalability:** The solution should support additional users and records.
 - **Usability:** Authorized users should be able to perform permitted operations easily.
 - **Performance:** Access checks should be performed efficiently.
 - **Data Integrity:** Unauthorized users should not be able to modify or delete protected records.
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4. User Requirements

The project contains different types of users:

User	Permission
Administrator	Full access
EEE User with bb1	Read permitted EEE records
User with bb2	Create records
User with bb3	Write/Edit records
User with bb4	Delete records
Unauthorized User	No access to restricted records

5. System Requirements

- Software Requirements
 - ServiceNow instance
 - Web browser
 - Internet connection
 - ServiceNow System Administrator access
 - Hardware Requirements
 - Computer/Laptop
 - Minimum 4 GB RAM
 - Internet connectivity
 - Keyboard and mouse
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6. Database/Table Requirements

A custom table named Institution Details will be created.

Table Name: u_institution_details

The table will contain the following fields:

Field	Type
Student Roll Number	Auto Number
Student Name	Reference – User
Faculty Name	Reference – User
Branch	Choice
Email	String
Phone Number	String
Description	Multi String

The Branch field will contain:

- ECE

- EEE
 - CSE
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7. ACL Requirements

The project will implement four main ACL operations:

ACL Operation	Required Role
Read	bb1
Create	bb2
Write	bb3
Delete	bb4

For the Read ACL, the Data Condition will restrict access to records where:

Branch = EEE

Administrators will have full access.

8. Security Requirement

The system must ensure that users cannot access records beyond their assigned permissions.

For example:

- EEE User + bb1 → EEE records
- User without required role → Access denied
- Admin → All records

This provides both role-based access control and field-based record restriction.

9. Expected Result

After fulfilling the requirements, the system should successfully:

- Protect institutional records.
 - Restrict access based on branch.
 - Control CRUD operations using roles.
 - Prevent unauthorized access.
 - Allow administrators to manage all records.
 - Demonstrate secure ACL implementation in ServiceNow.
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10. Conclusion

The Requirement Analysis phase defines the functional, non-functional, user, system, database, and security requirements of the project. These requirements will serve as the foundation for designing and developing the Script-Controlled ACL solution in ServiceNow.
